Akoizo Original Practice - Form 2

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Answer: _____

2. (2 points) XOR(1,1) equals…

A. 2

B. Undefined

C. 0

D. 1

Answer: _____

3. (2 points) Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only for any red LED

B. Only at 5 V

C. No

D. Always

Answer: _____

4. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Answer: _____

5. (2 points) NOT(A AND B) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. (NOT A) AND B

B. A AND B

C. (NOT A) OR (NOT B)

D. A OR B

Answer: _____

6. (2 points) 1.5 Ah / 0.30 A equals…

A. 5 J

B. 0.20 h

C. 5 h

D. 0.45 h

Answer: _____

7. (2 points) At fixed R, doubling V changes power by…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. A factor of one half

B. No change

C. A factor of four

D. A factor of two

Answer: _____

8. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Stores a clocked bit by definition

B. Measures resistance

C. Selects an input

D. Always adds inputs

Answer: _____

9. (2 points) Useful troubleshooting changes…

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only the answer label

B. All components randomly

C. One suspected factor at a time

D. Every wire at once

Answer: _____

10. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

11. (4 points) Give half-adder equations.

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

12. (4 points) Why separate branch resistors?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

Answer: _____
_____
_____

13. (4 points) Convert 2 Wh to joules.

Answer: _____
_____
_____

14. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

15. (4 points) Why can real charging exceed the ideal time?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

Answer: _____
_____
_____

16. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

17. (4 points) Explain a nodal branch current.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

18. (2 points) Define "Open circuit" in the context of Circuit Lab.

Answer: _____
_____
_____